

10 in an off-screen buffer. When you tell pygame to update, it takes the contents of this off-screen buffer and puts them in the real window.

12. Slow things down a bit.

Computers are very fast, and if the loop continued to the next iteration right away without pausing, the program might run faster than the designated frame rate. The line in this section tells pygame to wait until a given amount of time has elapsed in order to make the frames of our program run at the frame rate that we specified. This is important to ensure the program runs at a consistent rate, independent of the speed of the computer on which it's running.

When you run this program, the program just puts up a blank window filled with black. To end the program, click on the close button in the title bar.

Drawing an Image

Let's draw something in the window. There are two parts to showing a graphic image: first we load the image into the computer's memory, then we display the image in the application window.

With pygame, all images (and sounds) need to be kept in files external to your code. Pygame supports many standard graphic file formats, including .png, .jpg, and .gif. In this program we'll load a picture of a ball from the file *ball.png*. As a reminder, the code and assets associated with all the major listings in this book are available for download at <https://www.nostarch.com/objectorientedpython/> and <https://github.com/IrvKalb/Object-Oriented-Python-Code/>.

While we only need one graphic file in this program, it's a good idea to use a consistent approach to handling graphic and sound files, so I'll lay one out for you here. First, create a project folder. Place your main program in that folder, along with any related files containing Python classes and functions. Then, inside the project folder, create an *images* folder into which you'll place any image files you want to use in your program. Also create a *sounds* folder and place any sound files you want to use there. [Figure 5-6](#) shows the suggested structure. All of the example programs in this book will use this project folder layout.